

A Firm-Level Microsimulation for VAT Policy Analysis^{*}

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Preliminary and Incomplete. Comments are welcome.

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Abstract

I present an open, reproducible firm-level microsimulation for costing UK VAT registration-threshold reforms, released with its synthetic-data generator. The threshold is a *notch*: once turnover crosses £85,000, 20% VAT falls due on a firm’s whole turnover, not the excess. I build synthetic firm records from ONS counts, calibrate them to 2023–24 HMRC statistics, and cost *level*, *shape* (a taper) and *rate* (a reduced-rate band) reforms statically on a common basis, reproducing HMRC’s £85,000-to-£90,000 costing within £10m. I characterise the notch by its *dominated region*—the exact £21,250 band above the threshold in which no firm locates—and how each reform changes it: raising it (to £100,000, –£508m) relocates it, a reduced rate shrinks it, a taper removes it. I embed the behavioural layer—the registration decision—in the framework of Liu et al. (2021), in which a firm weighs the input-VAT reclaim benefit against the output-VAT burden on consumer sales, and price each reform’s budget cost on the registration margin parametrically over the input share and the business-to-consumer share, as a sensitivity range. A placebo shows the synthetic bunching is a calibration artefact, not behaviour.

Keywords: value-added tax, microsimulation, tax notch, policy

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1 Introduction

The United Kingdom’s Value Added Tax (VAT) applies above a registration threshold that is among the highest in the OECD (Seely, 2024; HM Revenue and Customs, 2024). The threshold is a *notch* rather than a kink: once a firm’s annual taxable turnover crosses it, the standard 20% rate falls due on all of the firm’s turnover, not only on the part above the threshold, so registration produces a discrete jump in liability rather than a marginal one. The threshold falls in a dense part of the firm distribution. Figure 1(a) shows that UK firms are concentrated at the lower end of the turnover distribution, and Figure 1(b) shows that VAT-registered firms are concentrated in the turnover bands at and below the threshold, so the notch applies to a large number of firms.

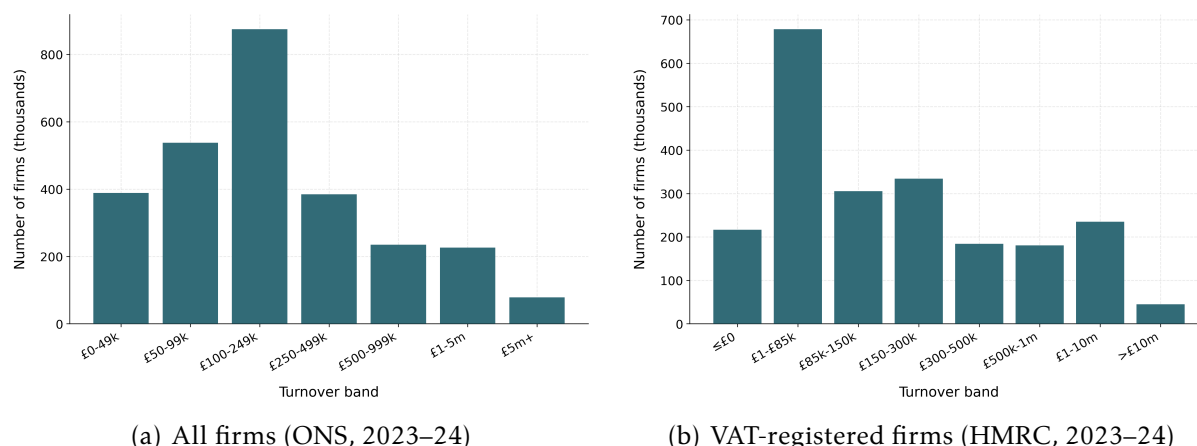


Figure 1: The UK firm-size distribution and VAT registration, 2023–24 (threshold £85,000). Panel (a): number of firms by turnover band. Panel (b): number of VAT-registered firms by turnover band. Source: panel (a), ONS, *UK Business: Activity, Size and Location* (firm counts by turnover band); panel (b), HMRC, *Value Added Tax (VAT) Annual Statistics* (VAT-registered population by turnover band).

Firms respond to the notch by locating below the threshold. Tabulations of HMRC turnover data show the number of firms rising towards £84,000 and then dropping sharply at the £85,000 threshold, leaving excess mass just below it (Office for Budget Responsibility, 2023). Liu et al. (2021) document this bunching in UK firm data, and Liu et al. (2024) find that firms slow their turnover growth as they approach the threshold. The notch therefore distorts the size distribution and the growth decisions of firms near the margin, which places the threshold under recurrent pressure for reform.

Reform proposals change the threshold in one of three ways: its *level* (raising or lowering it), its *shape* (replacing the notch with a schedule that phases the rate in over a band), or its *rate* (a reduced rate for firms in a band above the threshold). The three are not equivalent. Changing the level moves the notch along the turnover axis; changing the shape or the rate alters the size of the discrete jump itself. Costing the three on a

common basis is the practical difficulty this paper addresses. A change in the level can be costed by mechanical accounting, but a change in the shape of the schedule cannot be evaluated with the reduced-form bunching methods used to study the threshold, because a continuous schedule leaves no single point of excess mass to exploit. The firm-level data needed to study turnover near the threshold is, in addition, not available in open form.

In this paper I build an open, reproducible firm-level microsimulation for costing UK VAT threshold reforms on a common static basis. I construct synthetic firm microdata calibrated to published HMRC aggregates—band-level firm counts and sector turnover totals—for the 2023–24 tax year, when the threshold was £85,000 (raised to £90,000 in April 2024); the synthetic data can be released openly and regenerated under any counterfactual schedule. On these data I do four things. I cost both level reforms and schedule reforms—a graduated taper and a reduced-rate band—statically, by applying each counterfactual schedule mechanically to the firm distribution. I show that the synthetic data reproduce the density step at the £85,000 notch documented in administrative data by [Liu et al. \(2021\)](#), and I use a placebo test to demonstrate that this step is mechanical—inherited from the coarse band-level counts that serve as calibration targets rather than generated by firm location choice—so I draw no behavioural elasticity from the synthetic bunching. And I characterise the notch’s distortion through its *dominated region*—the exact, elasticity-free range of turnover above the threshold in which no firm locates—and how each reform changes it. I then additionally embed the notch in the registration framework of [Liu et al. \(2021\)](#)—in which registering brings an input-VAT reclaim against the output VAT due on business-to-consumer (B2C) sales—and price each reform’s budget cost on the registration margin, parametrically over the input share ω and the B2C share β . The static cost and the exact dominated region remain the elasticity-free robust anchors.

Three findings follow. First, the static pipeline prices level and shape reforms on a common basis and reproduces HMRC’s published anchor costing: my static costing of the April 2024 increase from £85,000 to £90,000 gives £175m in 2025–26 against the Government’s published £185m, agreeing within about £10m—an internal-consistency check on the costing rather than an external validation. Forward moves from the £90,000 base range from a gain of £665m at £70,000 to a cost of £1,205m at £120,000, against a base of about £203.8bn.

Second, the shape and rate reforms act on the notch’s distortion in a way that a change in level does not. The notch creates a *dominated region* of width $a = T^*\tau/(1-\tau) = £21,250$ above the threshold (£85,000–£106,250), a range of turnover in which no firm can locate without being strictly worse off than at the threshold. This region is an exact, elasticity-free property of the schedule (conditional on the turnover-tax notch): it follows from the arithmetic of the notch alone and requires no behavioural calibration. It also spans a populous part of the firm distribution—about 137,000 weighted firms—so its width is an economically meaningful distortion index, not a theoretical curiosity. Raising the threshold relocates this region without removing it. A reduced rate does *not* remove the distortion: it lowers the largest single notch (the jump at the threshold falls from

£17,000 to £12,750 at a 15% band) but, because the band reverts to the standard rate at its £105,000 top, it introduces a second notch and hence a second dominated region, so the *total* dominated-turnover width is essentially unchanged—£21,562 at 15% and £22,569 at 10%, both marginally above the £21,250 baseline. Only a graduated taper, being continuous, removes the distortion entirely. Set against their static revenue costs, the reforms are not interchangeable: raising the threshold to £100,000 costs £508m on this common £85,000 data-year base (the £90,000-base sweep of the first finding puts the same move at £374m), the most expensive option, yet removes none of the distortion; the taper removes the dominated region entirely for £336m; and the 10% and 15% reduced rates cost £343m and £171m while merely splitting the dominated band in two rather than shrinking it. The level move is therefore the dearest and least efficient of the menu. I report, for each reform, its static cost together with the portion of the dominated region it removes.

Pricing the same reforms on the registration margin makes the behavioural content the extensive-margin registration decision of [Liu et al. \(2021\)](#). Registering brings an input-VAT reclaim that rises in the input share ω , against an output-VAT burden on B2C sales that rises in the B2C share β ; the net incentive to register is proportional to $t[\omega - \beta(1 - \omega)]$, negative for consumer-facing, low-input firms—which therefore bunch below the threshold—and positive for input-heavy business-to-business firms, which register voluntarily regardless of where the threshold sits. Priced on this margin over (ω, β) —a registered firm yielding $t(1 - \omega)y$ and an unregistered firm $t\omega y$ —the reforms cost less than their static figures suggest. Raising the threshold to £100,000 costs between £0 and about −£1.05bn: revenue is lost only on low- ω , high- β consumer-facing firms, while input-heavy firms register voluntarily and so are unaffected by where the threshold is set. The 10% and 15% reduced-rate bands cost up to about −£0.93bn and −£0.47bn respectively. These figures are a parametric sensitivity range over (ω, β) , which are not grounded in the data, and capture the static extensive-margin registration response rather than the full CES-demand response of [Liu et al.](#)’s model; grounding (ω, β) in ONS Supply–Use data and computing the full CES response are left for future work. The static cost and the exact dominated region remain the elasticity-free anchors.

Third, the open data carry a transparency lesson about bunching. The synthetic records reproduce the density step at £85,000 that [Liu et al. \(2021\)](#) document in administrative data, which might tempt one to read a behavioural response off the synthetic distribution. A placebo test shows this would be a mistake: when the density step is removed from the calibration target, the apparent excess mass collapses from about 8,712 firms to roughly 99, revealing that the step is mechanical—a reflection of the two coarse turnover bands HMRC publishes across the threshold, baked into the calibration target—rather than a product of firm location choice, which the generator does not model. Behavioural bunching is thus not identified on the synthetic data, and I make no such claim from them; the behavioural evidence remains [Liu et al.](#)’s administrative estimate. I treat the reproduction of the published step as a credibility check on the data and the placebo as the appropriate caveat against over-reading it.

Related literature. The paper draws on and contributes to four strands of work. The first is the methodology of bunching and notches (Saez, 2010; Kleven and Waseem, 2013; Chetty et al., 2011; Kleven, 2016; Best et al., 2015), which provides the analytic notion of a dominated region; I use the dominated region to characterise a distortion that leaves no single point of excess mass to exploit. The second is the empirical literature on the VAT registration threshold (Liu et al., 2021, 2024; Onji, 2009; Harju et al., 2019; Asatryan and Peichl, 2017; Nandi and Warwick, 2020; Bashir et al., 2024), which estimates the behavioural response and studies changes in the threshold’s level. Liu et al. (2021) is closest: it estimates bunching at the UK notch on administrative data and documents voluntary registration; I take its administrative estimate as the established behavioural fact, show that my open synthetic data reproduce the density step it documents, and cost shape and rate reforms statically on a common basis. The international evidence (Harju et al., 2019; Onji, 2009; Asatryan and Peichl, 2017; Nandi and Warwick, 2020) notes that bunching at a VAT threshold can reflect compliance costs, firm-splitting, or reporting rather than real turnover responses. The third strand questions whether bunching identifies a structural elasticity at all: Blomquist et al. (2021) show that the elasticity is not non-parametrically identified from bunching at a kink without strong functional-form assumptions, and Bertanha et al. (2023) make the analogous point for notches. This literature motivates my refusal to read a behavioural elasticity off the synthetic bunching: even setting aside the placebo evidence that the synthetic step is mechanical, the mapping from excess mass to an elasticity is not identified without assumptions I am unwilling to impose. The fourth strand is the structural literature on size-based thresholds (Garicano et al., 2016; Gourio and Roys, 2014), which identifies policy-invariant primitives at the French 50-employee threshold and computes welfare from them; the present paper makes no such structural claim, pricing reforms statically and, conditional on an assumed elasticity, behaviourally. This paper makes three contributions. First, it provides an open, reproducible firm-level microsimulation for costing UK VAT threshold reforms, where the existing firm-level evidence on the threshold rests on restricted administrative data; as part of this, I show transparently that the open data reproduce the published density step and use a placebo to establish that the step is mechanical, so no behavioural bunching is claimed from the synthetic data. Second, it characterises the notch’s distortion through its exact, elasticity-free dominated region and costs two schedule reforms—a graduated taper and a reduced-rate band—on the same static basis as level reforms, where prior work has neither costed these reforms nor measured this distortion. Third, it embeds the registration decision in the framework of Liu et al. (2021)—the input-VAT reclaim and the B2C output-VAT burden—and prices each reform’s budget cost on this registration margin, parametrically over the input share and the B2C share, leaving the full CES-demand behavioural response and data-grounded (ω, β) for future work. A welfare and marginal-value-of-public-funds accounting—together with revenue-neutral designs and the treatment of voluntary registration—are also left for later.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background of the UK VAT threshold. Section 3 sets out the synthetic firm microdata

and its calibration to HMRC aggregates. Section 4 sets out the static costing of threshold changes and the schedule reforms. Section 5 reports the reduced-form bunching evidence. Section 6 develops the notch and its exact dominated region, and shows how each reform changes it. Section 7 sets out the Liu et al. (2021) registration model and prices each reform’s budget cost on the registration margin, parametrically over (ω, β) . Section 8 concludes.

2 Institutional background

Registration and the notch. The United Kingdom levies a value-added tax at a standard rate of 20% on most goods and services. A business must register once its taxable turnover over any rolling twelve-month period exceeds the registration threshold, after which it charges VAT on its taxable sales and reclaims VAT paid on inputs. Because registration makes VAT due on the firm’s *entire* taxable turnover rather than only the amount above the cutoff, the threshold is a *notch*—a discrete jump in liability—and not a kink. The cost of crossing is largest for consumer-facing firms with limited pass-through and small reclaimable input shares, for whom the incentive is to restrain turnover and locate just below the threshold (Liu et al., 2021; Kleven and Waseem, 2013).

Value added, input reclaim, and voluntary registration. Real VAT is a tax on *value added*, not on turnover: a registered firm charges output VAT on its sales but reclaims input VAT on its purchases. A large share of below-threshold firms therefore register *voluntarily*—roughly 43% of eligible below-threshold firms (Liu et al., 2021), driven by input-VAT reclaim and business-to-business sales—and for them the incentive to bunch is weak. My one-input model taxes *turnover* rather than value added, so it is a *turnover-tax-notch approximation* for the consumer-facing, limited-reclaim firms for whom the bunching incentive is sharpest; it abstracts from input reclaim and voluntary registration. I treat this as a first-order scope restriction.

Threshold history. The registration threshold was held at £85,000 from 1 April 2017 to 31 March 2024—a seven-year nominal freeze, the longest in the history of the tax—before being raised to £90,000 on 1 April 2024, with the deregistration threshold rising from £83,000 to £88,000 (HM Revenue and Customs, 2024; Seely, 2024). Because nominal turnover grows with inflation and real activity, freezing the threshold lowers it in real terms and draws an increasing number of firms towards the registration margin; this fiscal-drag mechanism was a central motivation for the April 2024 increase. My data year is 2023–24, so the prevailing statutory threshold throughout my analysis is £85,000. Figure 2 shows the pattern in the turnover distribution: the number of firms rises towards £84,000 and falls sharply above £85,000, and the projected 2025–26 distribution is more sharply bunched than the 2019–20 outturn as fiscal drag draws more firms towards the frozen threshold.

The reform debate. The UK threshold is among the highest in the OECD, and there is debate over whether to raise it—to reduce the compliance burden on small firms—

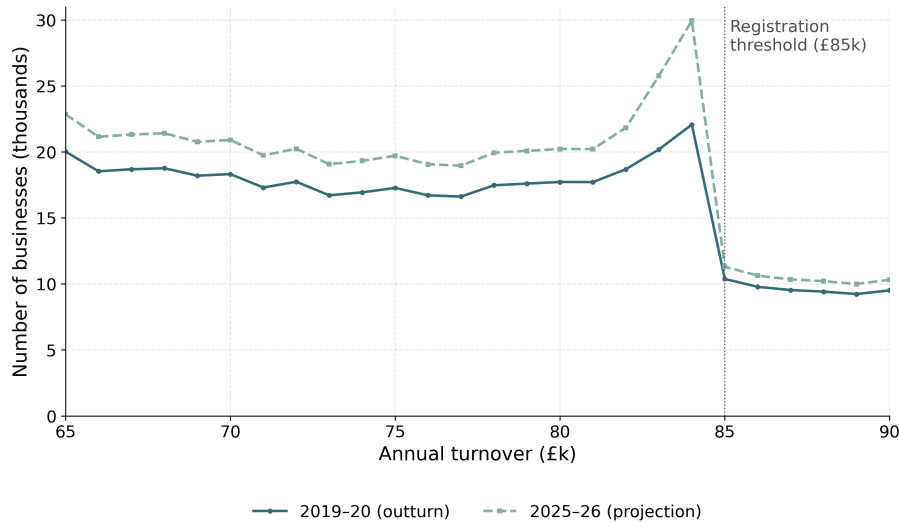


Figure 2: Bunching below the £85,000 VAT registration threshold: number of businesses by £1,000 turnover band. Source: [OBR, *Economic and Fiscal Outlook March 2023*](#), Chart C (Open Government Licence).

or lower it, to broaden the base and reduce the growth distortion at the margin. Three families of reform feature, and I evaluate all three: a change in the threshold’s *level* (the £85,000 to £90,000 increase of April 2024 and further moves around it); a change in its *shape* (a *graduated taper* that phases the effective rate in over a band, replacing the notch with a continuous profile); and a change in its *rate* (a *reduced rate* for small firms in a band above the threshold, with the standard 20% rate retained above). The taper and reduced-rate designs follow proposals in the UK policy debate, notably [Office of Tax Simplification \(2017\)](#), which canvassed a financial incentive or reduced effective rate to smooth the cliff-edge at the threshold. A banded reduced rate introduces a smaller *secondary* notch at the band top, which the continuous taper avoids. I cost all three by the same machinery in Section 4.

3 Data

Studying firm responses at the threshold needs firm-level turnover near the cutoff, which the official statistics publish only in coarse turnover bands. I therefore build a synthetic firm population calibrated to those aggregates: it reproduces the official totals while resolving the individual firm that the authorities do not release, opening the analysis to anyone. The same calibrate-to-published-aggregates approach underlies open household-level tax-benefit microsimulation ([PolicyEngine, 2025](#)); here I apply it at the firm level. The generator takes the registration threshold as a parameter and is documented in Appendix A.1.

The archived CSV inputs in this repository remain the reproduction source for the

paper’s reported numbers; a pinned migration snapshot also represents the same 2024–25 numeric target surface through PolicyEngine Ledger and the experimental Populace firm generator, and Appendix A.1 reports the parity check between that shared source-of-truth path and the paper inputs.

Construction. Index firms by i . For each sector s and ONS turnover band $b = [\underline{b}, \bar{b}]$, the *UK Business: Activity, Size and Location* table gives a firm count $N_{s,b}$, and I draw $N_{s,b}$ firms with within-band turnover

$$y_i = \underline{b} + (\bar{b} - \underline{b}) u_i + \varepsilon_i, \quad u_i \sim \mathcal{U}(0, 1), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_b^2),$$

truncated below at zero, where ε_i smooths the within-band distribution. Each firm is given input expenditure $x_i = \rho_i y_i$, with the input–output ratio ρ_i drawn from a rescaled Beta distribution with sector-specific shifts and bounded to $\rho_i \in [0.6, 1.5]$; its value added, the net VAT base, is $v_i = y_i - x_i$ (the upper bound $\rho_i = 1.5$ permits $x_i > y_i$, and hence negative value added, for a non-trivial share of firms in the calibrated population). Employment is assigned from the ONS employment-band shares.

The base population reproduces the ONS structure but not the HMRC VAT-registered totals, so I re-weight it. Each firm receives a strictly positive weight $w_i = e^{\theta_i}$, and for a vector of official targets $T = (T_k)_{k=1}^K$ the weighted synthetic counterpart of target k is

$$\hat{T}_k(\theta) = \sum_i a_{ki} w_i,$$

where a_{ki} is firm i ’s contribution to target k : an indicator of band or sector membership for a count target, and the firm’s liability v_i for a VAT-liability target. I choose θ to minimise a symmetric relative-error loss with per-target importance weights λ_k ,

$$L(\theta) = \frac{1}{K} \sum_k \lambda_k \min\left\{\left(\hat{T}_k/T_k - 1\right)^2, \left(T_k/\hat{T}_k - 1\right)^2\right\} + \frac{\alpha}{N} \sum_i |\theta_i|,$$

the final term a mean absolute penalty on the log-weights. The targets are the HMRC VAT-registered counts by turnover band and by sector, the HMRC net VAT liability by turnover band, and the ONS employment-band totals; turnover-band targets carry the largest λ_k , followed by VAT liability by band. VAT liability by sector is reported as an informational diagnostic but excluded from the optimizer, because the current generator fixes firm inputs rather than calibrating the input/output tax structure by sector. The problem is solved by gradient descent, and the symmetric loss keeps targets of very different magnitudes on a common scale.

A firm is registered with certainty when its turnover exceeds the threshold T^* . Below the threshold it registers voluntarily with probability $r = M/n_<$, where M is the HMRC count of registered firms in the £1–threshold band and $n_<$ is the unweighted number of synthetic below-threshold firm rows. This sets the expected row-level voluntary-registration rate; I do not treat below-threshold voluntary registration as a separate

weighted calibration target. The resulting dataset comprises approximately 2.94 million firm records, weighted to represent roughly 2.5 million UK firms; checked for consistency against the official sources along every calibrated dimension, it attains an overall calibration accuracy of about 90%. This headline is the mean of five calibrated-dimension scores, each clipped at zero as $\max\{0, 1 - |\hat{T} - T|/|T|\}$; it excludes the VAT-liability-by-sector diagnostic described above.

The registration threshold in the data. My data are for the 2023–24 year, in which the statutory threshold was £85,000. The firm density steps down at the threshold—from roughly 10,200 to 6,600 firms per £1,000 turnover band—reflecting the incentive to remain below the registration cutoff, with a second, smaller step at £150,000, the ceiling of the VAT Flat Rate Scheme (Figure 3). I note, however, that HMRC publishes firm counts only in coarse turnover bands across the threshold, so the density step at £85,000 in the synthetic data is inherited from these aggregate targets rather than from any modelled firm behaviour; Section 5 quantifies this with a placebo test. Counterfactual policies—including the £85,000 to £90,000 reform and the forward threshold sweep of Section 4—are evaluated on these firms by re-applying the registration rule to the microdata for the static costings. I hold the empirical turnover distribution fixed throughout. A fully behavioural forward solve that re-prices firms under counterfactual schedules is left to future work.

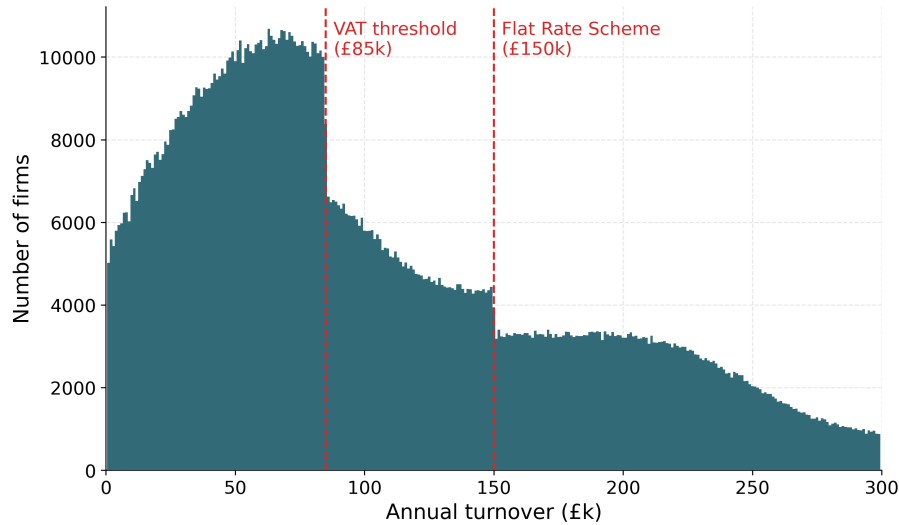


Figure 3: Synthetic firm turnover distribution (weighted firm counts, £1,000 bins), 2023–24. The density steps down at the £85,000 registration threshold and again at the £150,000 Flat Rate Scheme ceiling.

4 Static costing of threshold reforms

A static costing measures the first-order revenue effect of a reform by re-applying the tax rule to a fixed firm population, with no behavioural response. Let firm i have turnover y_i , calibrated net VAT liability v_i (output less input VAT; Section 3), and weight w_i . A firm remits VAT only once registered, that is once its turnover exceeds the threshold T , so total VAT revenue at threshold T is the weighted sum of liabilities over registered firms,

$$R(T) = \sum_i w_i v_i \mathbf{1}\{y_i \geq T\}.$$

Because turnover is held fixed, moving the threshold from T_0 to T changes revenue only through the firms whose registration status flips—those with turnover in the band between the two thresholds:

$$\Delta R = R(T) - R(T_0) = -\text{sign}(T - T_0) \sum_i w_i v_i \mathbf{1}\{T_{\min} \leq y_i < T_{\max}\},$$

where $T_{\min} = \min(T, T_0)$ and $T_{\max} = \max(T, T_0)$: lowering the threshold draws firms into the net and raises revenue, while raising it releases firms and loses revenue. The estimate is therefore the mechanical reclassification effect and excludes the behavioural response of Section 7. The calibrated weights reproduce HMRC’s net VAT-liability totals (Section 3), and I age the microdata to a given fiscal year by a cumulative nominal-growth factor. At the £90,000 threshold this puts total VAT revenue at £203.8bn in 2025–26 and £210.0bn in 2026–27.¹

Consistency check against HMRC. I first check the static pipeline against HMRC’s own costing of the April 2024 increase from £85,000 to £90,000, which was announced and costed at Spring Budget 2024 (HM Treasury, 2024). Because my microdata are calibrated to HMRC VAT aggregates (Section 3), a close match confirms internal consistency rather than out-of-sample accuracy. Figure 4 compares the two costings year by year: they share sign and profile throughout—a loss that narrows as fiscal drag lifts the frozen baseline towards the reformed threshold, turning to a small gain by 2028–29—and agree to within £7m–£21m per year (for 2025–26, –£175m against the published –£185m). The model reproduces both the magnitude and the turning point of the official costing.

A menu of threshold locations. Having checked the method, I use it to cost a menu of further threshold locations, taking the post-reform £90,000 threshold as the baseline. Each row of Table 1 costs a move of the threshold to a new location for 2025–26, holding the turnover distribution fixed, so the revenue change reflects only the reclassification

¹This £203.8bn aggregate is the weighted sum of the synthetic registered population’s net VAT liabilities (output less input VAT) aged to 2025–26—a theoretical-liability (VTTL-style) base, not the published cash-receipts line. It therefore sits above published net cash VAT receipts (around £169bn in 2023–24, and an OBR forecast near £180bn for 2025–26), which are reduced relative to the theoretical liability by refunds and repayments, the VAT gap, and collection timing.

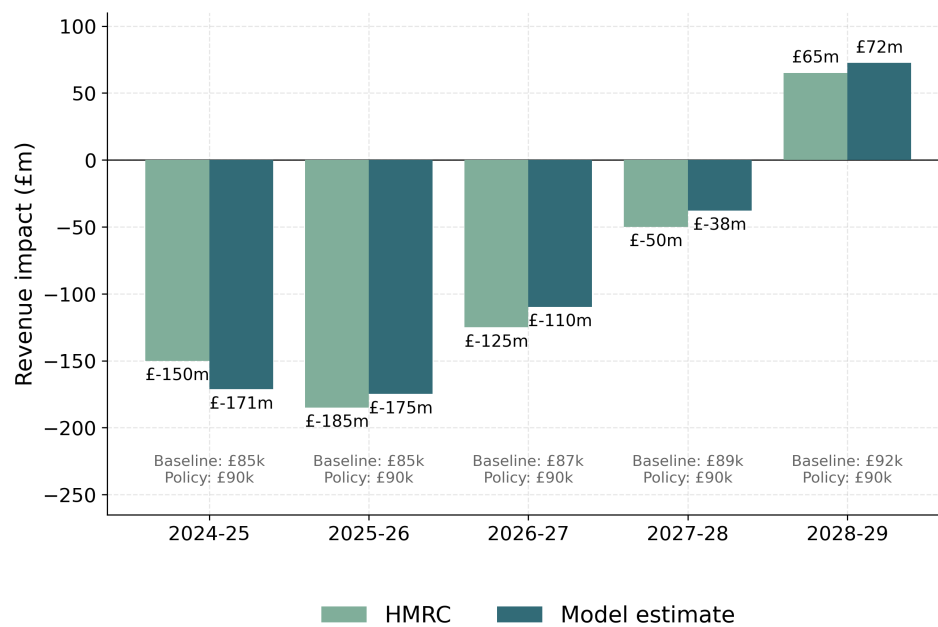


Figure 4: Anchor reform: revenue impact of raising the VAT registration threshold from £85,000 to £90,000 (the April 2024 change), my estimate versus HMRC’s published costing, 2024–25 to 2028–29. The two series share sign and profile, both turning positive in 2028–29 as fiscal drag lifts the frozen baseline above the reformed threshold. Because the microdata are calibrated to HMRC aggregates, this is a consistency check, not an out-of-sample validation.

of firms into or out of the VAT net. Because the calibration places extra weight on high-liability firms just below the threshold, I evaluate the reclassified band from a smooth extrapolation of the clean above-threshold profile rather than the raw below-threshold mass. Lowering the threshold draws firms in and raises revenue; raising it loses revenue. The schedule is approximately linear, with a mild convexity driven by the higher individual VAT liabilities of higher-turnover firms—each £5,000 step changes revenue by roughly £150m–£220m, and the marginal revenue per step grows (£183.5m $\rightarrow$ £218.1m as the threshold rises) even though the number of firms reclassified per step falls (from about 18,600 to about 16,200), because each firm released at a higher threshold remits more VAT. Even a £30,000 rise to £120,000 costs about £1.2bn, under 0.6% of the £203.8bn base, because the threshold governs many small firms whose individual liabilities are small. Figure 5 shows the sweep graphically. These are *static* results: they bound the first-order fiscal effect of a threshold change but abstract from the behavioural response of Section 7.

Table 1: Static revenue and VAT-paying-firm effects of moving the VAT registration threshold, 2025–26, relative to the post-reform £90,000 baseline (total VAT revenue £203.8bn). Figures are mechanical, holding firm turnover fixed (no behavioural response). This sweep is distinct from the anchor reform of Figure 4.

Threshold	Revenue change (£m)	Change (%)	VAT-paying firms (000s)
£70,000	+664.9	+0.33	+79.3
£75,000	+509.0	+0.25	+58.7
£80,000	+346.3	+0.17	+38.7
£85,000	+176.6	+0.09	+19.1
£90,000 (baseline)	0	0	0
£95,000	−183.5	−0.09	−18.6
£100,000	−374.0	−0.18	−36.8
£105,000	−571.4	−0.28	−54.5
£110,000	−775.7	−0.38	−71.7
£115,000	−987.0	−0.48	−88.5
£120,000	−1,205.1	−0.59	−104.7

4.1 Costing schedule reforms

The threshold sweep above moves the *location* of a single cutoff. The same static method—re-applying a tax rule to the fixed turnover distribution, with no behavioural response—also costs reforms that change the *shape* of the schedule rather than merely its position. I consider four reforms and report their first-order revenue effects against a common baseline: the £85,000-notch / £183.6bn 2023–24 base. (This 2023–24 data-year base differs from the £203.8bn figure in the sweep above, which ages the same microdata

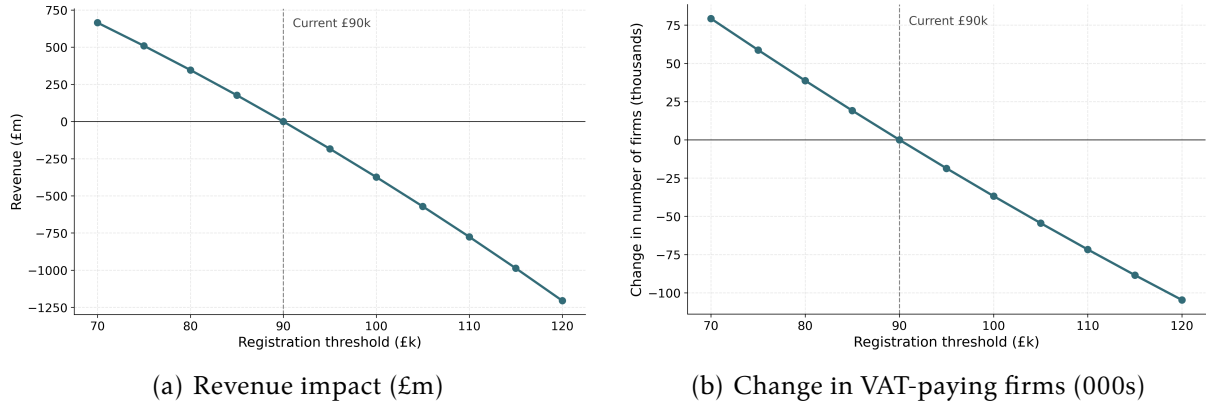


Figure 5: Static effect of moving the VAT registration threshold, 2025–26, relative to the £90,000 baseline (dashed line). Panel (a): revenue impact—lowering the threshold raises revenue, raising it loses revenue, and the schedule is approximately linear, with a mild convexity driven by the higher individual VAT liabilities of higher-turnover firms—the marginal revenue per £5,000 step grows even as the number of firms reclassified per step falls—over the £70,000–£120,000 range. Panel (b): the change in the number of VAT-paying firms, tracking the local density of firms on each side of the threshold.

to the 2025–26 fiscal year at the post-reform £90,000 threshold; the schedule reforms are costed in the data year to match the bunching and dominated-region analysis. Both £183.6bn and £203.8bn are theoretical-liability (VTTL-style) bases in the sense of the footnote above, not cash-receipts lines.) For each reform I recompute the weighted sum of liabilities $R = \sum_i w_i v_i$ under the new schedule, holding each firm’s turnover fixed, and difference it against the baseline.

Threshold relocation. Raising the threshold from £85,000 to £100,000 is a pure location move, costed exactly as in the sweep above: it releases the firms with turnover in $[\text{£}85,000, \text{£}100,000)$ —about 94,300 firms—from the VAT net, for a static revenue change of $-\text{£}508\text{m}$ on the £85,000-notch / £183.6bn base, consistent (same base) with the taper and reduced-rate reforms below. On this common base the level move is the most expensive option yet removes no distortion—it merely relocates the dominated region to a new cutoff—whereas the taper removes the dominated region entirely at lower cost ($-\text{£}336\text{m}$), foreshadowing Section 6.

Graduated taper. A graduated taper replaces the discrete notch with an effective rate that phases linearly from 0% at £85,000 to the full 20% at £105,000. Firms in the band remit a tapered rather than a full rate on their (fixed) turnover; mechanically this reduces revenue by $-\text{£}336\text{m}$, with about 123,100 firms in the £85,000–£105,000 band facing a lower effective rate.

Banded reduced rate. A banded reduced rate charges firms in the £85,000–£105,000 band a reduced rate $\tau_{\text{low}} \in \{10\%, 15\%\}$ in place of the standard 20%, with 20% re-

tained above £105,000. Statically this reduces revenue by –£343m at $\tau_{\text{low}} = 10\%$ and by –£171m at $\tau_{\text{low}} = 15\%$, again with about 123,100 firms in the band facing the lower rate.

Table 2: Static, behaviour-free revenue effect of the four reforms relative to the common £85,000-notch / £183.6bn 2023–24 baseline. Each reform is costed mechanically by re-applying its schedule to the fixed turnover distribution, with no behavioural response.

Reform	Lever	Static revenue (£m)
Raise threshold to £100,000	Location	–508
Graduated taper [£85k, £105k]	Shape (phase-in)	–336
Reduced rate 10% [£85k, £105k]	Rate (step)	–343
Reduced rate 15% [£85k, £105k]	Rate (step)	–171

These static figures bound the first-order fiscal effect of each reform but abstract from the behavioural response of Section 7. One feature of the shape reforms is not visible in a static costing: unlike a threshold move, which simply transports the notch to a new cutoff, the taper and the reduced rate *change the size of the notch itself*—the discrete drop in net revenue at the threshold and the empty band of turnover it induces. Quantifying that change requires the structural object I develop in Section 6, where I derive the notch’s dominated region and show analytically how each of these reforms alters it.

5 Bunching at the VAT threshold

This section applies the polynomial counterfactual-density estimator of Saez (2010) and Kleven and Waseem (2013)—as used on UK VAT data by Liu et al. (2021)—to my synthetic firm population at the £85,000 registration threshold prevailing in my 2023–24 data. The estimator is purely reduced-form: it fits a smooth counterfactual to the empirical turnover density and reads excess mass off the gap, without imposing any structural model of the firm. My headline finding is one about my data rather than about firm behaviour. The estimator returns a sizeable excess-mass spike at the threshold, but a placebo test shows that this spike is *mechanical*: it is inherited from the coarse turnover-band targets I calibrate to, not generated by any location-choice in the synthetic firms. I therefore report the estimate as a reproduction of a published density step, stress-test it, and make no behavioural bunching claim and convert no excess mass into an elasticity. I read this as a transparency result: the open synthetic file faithfully transmits the administrative pattern, and its limit is reported openly.

5.1 Methodology

Following Liu et al. (2021), I group firms into fine turnover bins of £1,000 and estimate a counterfactual density—the distribution that would obtain absent the registra-

tion notch—by polynomial regression on bin counts, excluding a manipulation window around the threshold y^* :

$$c_j = \sum_{l=0}^q \beta_l (y_j)^l + \sum_{i \in [y_-^*, y_+^*]} \gamma_i 1\{j = i\} + \varepsilon_j,$$

where c_j is the number of firms in bin j , y_j is the distance of bin j from the threshold, q is the polynomial order, and $[y_-^*, y_+^*]$ is the excluded window. The fitted polynomial is then projected into the excluded region to recover the counterfactual density $g(y)$. Excess bunching is the integrated difference between the observed and counterfactual densities just below the threshold,

$$B(y^*) = \int_{y^* - \Delta y^*}^{y^*} g(y) dy \approx g(y^*) \Delta y^*,$$

the approximation holding when g is locally smooth. The bunching ratio $b(y^*) = B(y^*)/g(y^*) \approx \Delta y^*/y^*$ is the width of the bunching region expressed as a fraction of the threshold—a normalised *ratio*, not an elasticity. The estimator delivers the raw geometry of any density step at the threshold; whether that step reflects firm behaviour is a separate question, addressed by the placebo test below.

5.2 Results

Applied to the 2023–24 synthetic data at the £85,000 threshold, the estimator returns a bunching ratio $b = 0.060$ (SE 0.005) and excess mass of 8,712 firms (SE 662), with full inference—bootstrap standard errors and the mass-conservation restriction that excess mass below the threshold equal missing mass above it—reported in Appendix A.2 and reproducible from the released code on the regenerated synthetic data. The observed density displays a clear step at the threshold: more mass in the £1,000 bins immediately below the cutoff than just above it.

I want to be precise about what this number is. It is the magnitude of a density step in my synthetic file, and that step reproduces the discontinuity documented in UK administrative micro-data by Liu et al. (2021): the open synthetic population transmits the published pattern. It is *not* a measured behavioural response. The synthetic generator contains no firm location-choice, so the firms in my file do not—and cannot—adjust their turnover to avoid registration. Whether the reproduced step nonetheless carries any behavioural information is exactly what the placebo test settles, and the answer is that it does not.

5.3 Placebo test

The central result of this section is a placebo test that diagnoses where the excess mass comes from. The test exploits a feature of my calibration targets. HMRC publishes

turnover counts only in coarse bands across the threshold: 678,350 firms at or below £85,000 (an average of about 7,981 firms per £1,000) and 305,320 between £85,000 and £150,000 (about 4,697 per £1,000). The two band densities differ, so a step at £85,000 is already present in the calibration target before any firm is generated. Because the generator has no location-choice, any density discontinuity it produces must be inherited from this target rather than chosen by firms.

I therefore remove the step from the target in two ways and re-run the estimator. In Placebo A I reweight the existing synthetic firms so that the per-£1,000 density follows a single smooth trend across the threshold, holding total mass fixed. In Placebo B I replace the two HMRC band counts with a split implied by one common density, then regenerate the entire population and re-calibrate from scratch. The full table is in Appendix A.2. The result is decisive: excess mass collapses from 8,712 firms to 0 under Placebo A and to 99 under Placebo B—about one per cent of the original spike—and the bunching ratio turns slightly negative in both. Removing the step from the calibration target removes essentially all of the bunching.

The conclusion I draw is plain. The synthetic bunching is inherited from the coarse HMRC band targets, not generated by firm behaviour, because the generator has no location-choice. I therefore make no behavioural bunching claim on these data, and I do not convert the excess mass into a turnover elasticity. The apparent agreement between my normalised statistic and LLAT's (see below) is a mechanical coincidence fixed by the band-density ratio, not an independent behavioural match. This is a credibility result more than a negative one: the open synthetic file is stress-tested and its limit is reported transparently. The exercise also sits comfortably with the broader identification critique of bunching designs, which warns that excess mass need not point-identify a structural elasticity even in genuine administrative data (Blomquist et al., 2021; Bertanha et al., 2023); here the warning is sharper still, since the mass is not behavioural to begin with. A recovery test in Appendix A.2 confirms the null is a property of the data rather than the estimator: when a behavioural signal of known magnitude is injected into the step-free population, the same estimator recovers about 90% of it, so it is both specific (the placebo) and sensitive (the recovery).

5.4 Comparison with the literature

The behavioural fact in the literature is administrative. Liu et al. (2021) (hereafter LLAT) document genuine bunching at the same £85,000 UK VAT notch in HMRC micro-data, over a manipulation window of −£14k to +£24k, with an excess-bunching ratio of $b = 1.361$ (SE 0.202). If I normalise my synthetic excess mass to the same statistic I obtain $b_{LLAT} = 0.94$, of similar order of magnitude. I stress that this proximity is not an independent behavioural confirmation. As the placebo test above shows, my number is a mechanical reflection of the band-density ratio in the HMRC targets; its closeness to LLAT's behavioural estimate is a coincidence of that ratio rather than a second measurement of the same response.

What does carry across is a piece of institutional context that both settings share. Like

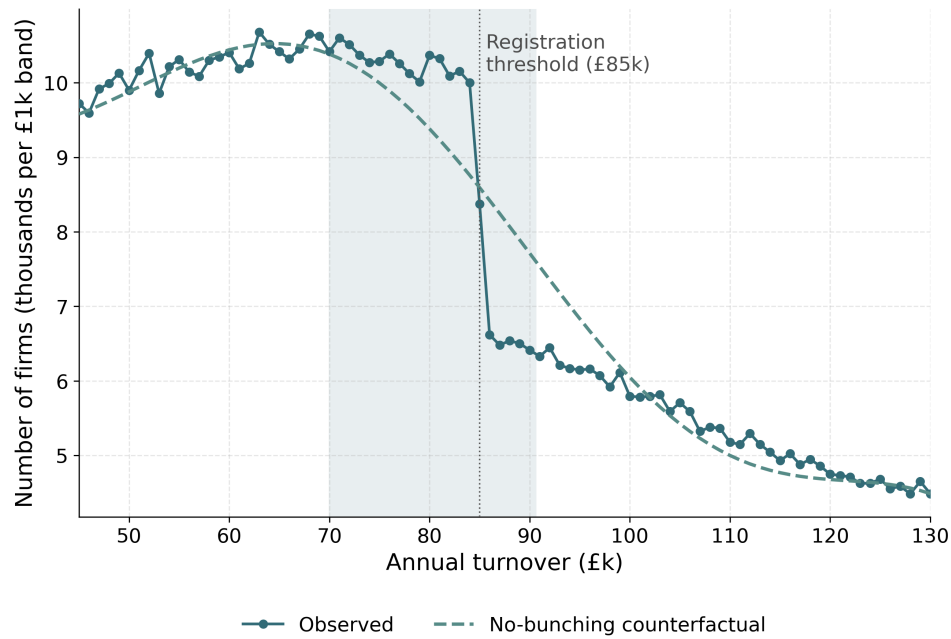


Figure 6: Reproduced density step at the £85,000 threshold (2023–24 data year). The figure shows the observed firm density in the synthetic population and the polynomial no-bunching counterfactual fitted outside the manipulation window. The synthetic data reproduce the density step documented in UK administrative micro-data by [Liu et al. \(2021\)](#); the gap below the threshold has magnitude $b = 0.060$ (a width ratio) and $E = 8,712$ firms. These describe the size of the reproduced step, not a behavioural estimate: the placebo test in Appendix [A.2](#) shows the step is inherited from the coarse calibration targets rather than from firm behaviour.

LLAT, the UK market features substantial voluntary registration below the threshold—LLAT report that almost half (roughly 43%) of eligible below-threshold firms register voluntarily—which dampens observed bunching relative to a pure turnover-tax notch and which my turnover-tax-notch model abstracts from. For wider order-of-magnitude context, [Office for Budget Responsibility \(2023\)](#) count firms holding turnover below the frozen threshold over a *wide* band (rising from 23,000 in 2017–18 to ~ 44,000 in 2025–26, with foregone turnover of £110m rising to £350m); this is a different object from a near-threshold spike and is cited only as corroborating context, not as a like-for-like match.

5.5 Reduced-form evidence and the dominated region

The placebo sharpens, rather than weakens, the case for grounding the policy analysis in structure. The reduced-form density step cannot price reforms—it has no single point of excess mass to forward-map onto a change in the *shape* of the schedule—and on these data it cannot even be read behaviourally at all. What the structure of Section 6 *does* deliver exactly, for every reform considered, is the notch’s dominated region and how each schedule change alters it. It is precisely because the reduced-form bunching is non-identifying here that the exact dominated region carries the distortion characterisation. A fully behavioural forward solve that re-prices each firm’s turnover under counterfactual schedules would require further parameter-sensitive assumptions and is left to future work.

6 The notch and its dominated region

The behavioural response to the registration notch is documented in administrative data by [Liu et al. \(2021\)](#); on my synthetic data the reduced-form bunching is not behaviourally identified (Section 5). To characterise the notch precisely I therefore turn to its analytic structure, which yields an exact distortion measure that needs no behavioural calibration. I set out the firm-level structure following the framework of [Kleven and Waseem \(2013\)](#), [Saez \(2010\)](#), and [Kleven \(2016\)](#), as applied to UK VAT by [Liu et al. \(2021\)](#). The UK registration threshold is a *notch*, not a kink: once turnover crosses the threshold the firm becomes liable for VAT on its *entire* turnover, not merely on the increment above it. I model this notch exactly. The section’s central deliverable is the notch’s *dominated region*—an exact, elasticity-free accounting identity that indexes the distortion the threshold imposes (Section 6.2), whose analytic response to each schedule reform complements the static reform costs of Section 4.1. The dominated region is a transparent, parameter-free distortion index: it depends on the rate and threshold alone and is therefore exact. Section 7 prices each reform’s budget cost on the registration margin of [Liu et al. \(2021\)](#), parametrically over the firm’s input and business-to-consumer shares; here the dominated region remains the exact, elasticity-free distortion measure, requiring no assumption about the firm’s technology, demand, or registration behaviour.

Scope: a turnover-tax notch. The structure here taxes *turnover*. Real VAT is levied on *value added* with input-tax reclaim, and a substantial share of below-threshold firms register voluntarily to recover input VAT (Liu et al., 2021). My treatment abstracts from input reclaim and voluntary registration: it is a turnover-tax-notch approximation. This is a scope limitation I state up front; it bounds the welfare claims I make and is one I return to in the concluding discussion.

6.1 The firm's problem

Each firm i chooses turnover y_i to maximise after-tax profit. Let $c_i(y_i)$ denote the cost of producing turnover y_i , T^* the registration threshold, and τ the standard VAT rate. Because VAT applies to the firm's entire turnover once it registers, after-tax revenue is discontinuous and profit is

$$\pi_i(y_i) = \begin{cases} y_i - c_i(y_i), & y_i < T^* \quad (\text{unregistered}), \\ (1 - \tau) y_i - c_i(y_i), & y_i \geq T^* \quad (\text{registered}). \end{cases} \quad (1)$$

At T^* after-tax revenue falls discretely by τT^* : a firm that nudges its turnover from just below to just above the threshold loses τT^* of net revenue on turnover it was already earning. With the standard rate $\tau = 0.20$ and threshold $T^* = \text{£}85,000$ this discrete drop is $\tau T^* = \text{£}17,000$. This drop—absent from a kink, where only the *marginal* rate changes at the threshold while average liability moves continuously—is the defining feature of a notch and the source of the bunching documented in Section 5. It is what makes locating just below the cutoff strictly attractive and creates the empty band derived in Section 6.2.

6.2 The dominated region

The notch creates a range of turnover just above T^* that *no* firm chooses, because every point in it yields strictly lower profit than locating at T^* . A firm at the threshold earns net revenue T^* ; a registered firm at $T^* + a$ earns $(1 - \tau)(T^* + a)$. Equating the two—the cost difference across this narrow band is negligible—the upper edge of the dominated region is

$$(1 - \tau)(T^* + a) = T^* \implies a = T^* \frac{\tau}{1 - \tau}. \quad (2)$$

This is the Kleven–Waseem dominated-region width (Kleven and Waseem, 2013). At my parameters it gives $a = \text{£}85,000 \times 0.20/0.80 = \text{£}21,250$, so the dominated region is $(T^*, T^* + a) = (\text{£}85,000, \text{£}106,250)$, shown in Figure 7. Two properties make this $\sim \text{£}21\text{k}$ band a transparent, parameter-free index of the distortion. First, it is *exact and elasticity-free*: it depends on τ and T^* alone and is independent of the cost specification, of any behavioural elasticity, and of any smoothing parameter. It is, however, *conditional on the statutory-rate turnover notch*: the $\text{£}21,250$ width is computed at the statutory $\tau = 0.20$ applied to whole turnover, and so applies to the limited-reclaim, consumer-facing firms that are the scope of this model (the “Scope: a turnover-tax notch” paragraph above).

For firms with substantial input reclaim the effective notch—and hence its dominated region—is correspondingly smaller. The width is thus independent of any behavioural elasticity but conditional on the turnover-tax-notch scope. Second, it is an order of magnitude wider than the $\sim \text{£}1\text{k}$ region implied by a smoothed *kink* approximation. The band also spans a populous part of the firm distribution: about 137,000 weighted firms in the synthetic population lie in the $\text{£}85,000\text{--}\text{£}106,250$ dominated band. Because the synthetic generator has no location choice, this counts the firms the data *place* in the band—indexing the size of the population exposed to the distortion—rather than firms that *optimally locate* there; the identity thus ranges over a heavily populated stretch of turnover rather than an empty tail. The precise count is, moreover, conditional on the uniform within-band fill used to resolve turnover inside the coarse HMRC bands (Section 3) and would shift under a different within-band shape; it is the width a , not this count, that is exact. The dominated region stands alone as a theoretical property of the schedule, independent of any reduced-form pattern in the data.

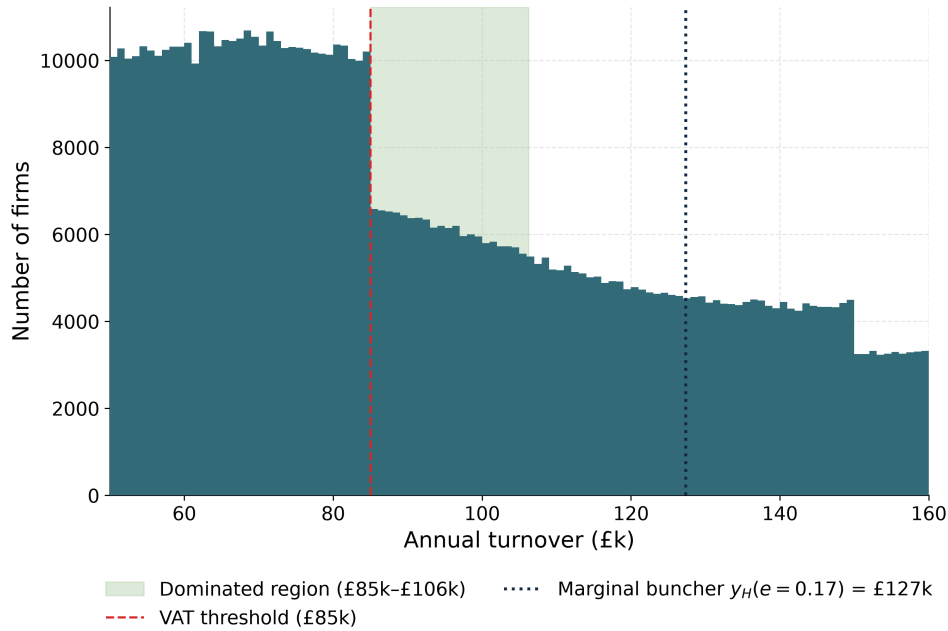


Figure 7: The $\text{£}85,000$ notch and its dominated region. The observed synthetic turnover density is shown with the dominated region ($\text{£}85,000, \text{£}106,250$) shaded; no firm optimally locates there. The marginal buncher at an assumed turnover elasticity $e = 0.17$ sits at $n_H = \text{£}127,382$ (Appendix A.2). The dominated-region width $a = \text{£}21,250$ is exact and independent of any elasticity.

Value-added robustness of the dominated region. The $\text{£}21,250$ width assumes VAT falls on the firm’s whole turnover, whereas real VAT taxes value added net of reclaimed input tax. A firm that remits a net rate τ_0 (its net VAT as a share of turnover) faces a value-added dominated region $a_i = T^* \tau_0 / (1 - \tau_0)$ rather than the full-rate width. I gauge the

magnitude this implies from the firm-level net-rate distribution for near-threshold firms (turnover £80k–£90k). About 42% of these firms have a net VAT rate below 1%—net input creditors or near-zero remitters—so they face effectively *no* notch; this share is close to the roughly 43% voluntary-registration rate [Liu et al. \(2021\)](#) document. For the remaining firms with positive net VAT, the value-added dominated region has a weighted median of about £18,650—the same order of magnitude as the £21,250 full-rate figure—rising to about £44,000 for the consumer-facing, high-net-rate firms in the top quartile, which are precisely the firms the turnover-tax-notch model is meant to describe. The value-added correction therefore does not collapse the dominated region: it identifies the $\sim 42\%$ of firms (the voluntary registrants) for whom there is no notch—the subpopulation already scoped out above—while leaving a $\sim £20k$ dominated region for the limited-reclaim, consumer-facing firms the model targets. The £21,250 width is thus a reasonable representative magnitude for that target population, consistent with the conditional, scope-limited reading set out above.

How each reform changes the dominated region. Because the width $a = T^*\tau/(1 - \tau)$ depends on the threshold and the rate alone, the effect of each schedule reform of Section 4.1 on this misallocation zone is exact and follows directly from the notch arithmetic of (2) and (3), with no dependence on the cost specification, any behavioural elasticity, or any simulated re-optimisation. Raising the threshold to £100,000 leaves the rate unchanged and therefore *relocates* the zone without altering its width: it is the same £21,250 band, transported to the new cutoff. A banded reduced rate does *not* remove the distortion. It reduces the *largest single* notch—the jump at T^* falls from $\tau T^* = £17,000$ to $r T^* = £12,750$ at a 15% band rate—but, because the band reverts to the standard rate τ at its upper edge $T_1 = £105,000$, it introduces a *secondary* notch there, and hence a second dominated region the simple “ a shrinks with τ ” reasoning omits. The primary region at T^* (the band rate r on whole turnover) has the usual width $a = T^*r/(1 - r)$: $£85,000 \times 0.15/0.85 = £15,000$ at 15% and $£85,000 \times 0.10/0.90 = £9,444$ at 10%. The secondary region at the band top—where the schedule steps from r back to τ on whole turnover—has width

$$a' = T_1 \frac{\tau - r}{1 - \tau}, \quad (3)$$

giving $£105,000 \times 0.05/0.80 = £6,562$ at 15% and $£105,000 \times 0.10/0.80 = £13,125$ at 10%. The *total* dominated-turnover width is therefore $a + a' = £21,562$ at 15% and $£22,569$ at 10%—both marginally *above* the £21,250 baseline, not below it. The reduced rate splits one wide empty band into two narrower ones rather than shrinking the total: it lowers the largest single notch (to £12,750 at 15%) at the price of adding a secondary notch of $(\tau - r)T_1 = £5,250$. The firm counts confirm this. The roughly 96,500 and 66,100 weighted firms previously reported for the 15% and 10% bands are the mass of the *primary* region alone; the secondary region above £105,000 restores comparable dominated mass, so the total dominated mass is not materially reduced. A graduated taper, by contrast, has *no* notch and hence no dominated region: marginal net revenue declines smoothly through the band rather than off a cliff, so the band of dominated turnover—and with it the discrete incentive to suppress turnover—is removed entirely. On the

dominated-region metric, then, *only* the taper removes the distortion; the reduced rate and the level move both leave it essentially intact—split and relocated, respectively—which sharpens the taper’s advantage. The dominated region is thus the paper’s exact, elasticity-free measure of the distortion the notch imposes, and the analytic statement above is exactly how much of that distortion each reform of Section 4.1 removes.

7 Behavioural reform costs: registration heterogeneity

The reforms of Section 4 are costed *statically*: each schedule is re-applied to a fixed turnover distribution, and the revenue change is the mechanical reclassification of firms with no response in turnover. Section 5 then shows that the bunching in my synthetic data is mechanical, so a behavioural elasticity is not identified from it—the reduced-form response is a placebo. Section 6 supplies the one object that needs no behavioural calibration at all: the exact, elasticity-free dominated region. What is missing from this sequence is the behavioural layer itself. In the UK VAT setting that layer is, in the first instance, not an intensive turnover response but an *extensive* one: the decision of whether to register. I follow Liu et al. (2021), who place the registration choice at the centre of the threshold notch and document its empirical signature directly in the administrative data—most strikingly, that roughly 43% of UK firms below the threshold register *voluntarily*. This section sets out their structure in full and uses it to price the threshold and rate reforms parametrically on the registration margin. Throughout, I am explicit that the elasticity-free anchors of the paper remain the static cost of Section 4 and the exact dominated region of Section 6; the registration costing here is a transparent, composition-conditional sensitivity layer that sits on top of them.

7.1 The firm’s problem: a registration model

I adopt the structural registration model of Liu et al. (2021), in the sufficient-statistics tradition of Saez (2010) and Kleven and Waseem (2013). Consider a small firm of productivity a and input share $\omega \in [0, 1]$ that sells to two markets: final consumers (business-to-consumer, B2C, quantity x) and other businesses (business-to-business, B2B, quantity y). Production is fixed-coefficients, following Keen and Mintz (2004): one unit of output requires ω/a units of a taxed intermediate input and $(1 - \omega)/a$ units of labour, with the input price and the wage both normalised to one. Absent VAT the unit cost is therefore $\omega/a + (1 - \omega)/a = 1/a$, and ω is the share of costs that are taxed, deductible inputs. The defining feature of VAT is that this input tax is reclaimable only by a registered firm, so the firm’s unit cost depends on its registration status. Writing the VAT rate as t , a registered firm reclaims the input VAT and so still pays ω/a for inputs, while an unregistered

firm pays the VAT-inclusive price $(\omega/a)(1+t)$; the unit cost is thus

$$c = \begin{cases} \frac{\omega}{a} + \frac{1-\omega}{a} = \frac{1}{a}, & \text{registered,} \\ \frac{\omega}{a}(1+t) + \frac{1-\omega}{a} = \frac{1+\omega t}{a}, & \text{unregistered.} \end{cases} \quad (4)$$

The gap $\omega t/a$ between the two is the irrecoverable input VAT an unregistered firm bears—the reclaim benefit of registering, rising in the input share ω , which is the first of the two forces in (8) below.

Demand in each market is iso-elastic (CES). Let $I \in \{0, 1\}$ be the registration indicator, e_C and e_B the B2C and B2B demand elasticities, A_C, A_B market scale terms, and λ the B2C taste weight. With p_C, p_B the firm's pre-tax prices,

$$\begin{aligned} x &= \lambda A_C (p_C(1+It))^{-e_C}, \\ y &= (1-\lambda) A_B p_B^{-e_B}. \end{aligned} \quad (5)$$

Output VAT grosses up the price the buyer faces only in the B2C market, and only when the firm is registered—this is the factor $1+It$ in the first line of (5). In the B2B market the buyer reclaims any VAT charged, so output VAT is neutral on that side and does not enter the second line. Let β denote the firm's B2C sales share. The firm sets a constant-elasticity monopoly markup over its unit cost in each market,

$$p_C = \frac{e_C}{e_C - 1} c, \quad p_B = \frac{e_B}{e_B - 1} c, \quad (6)$$

where c is the registration-dependent unit cost of (4). Because the consumer pays the grossed-up price $p_C(1+t)$ once the firm registers, registration simultaneously lowers the firm's unit cost (through input reclaim) and raises the price consumers face (through output VAT), so it depresses B2C quantity even as it expands the B2B margin. The firm chooses prices and the registration indicator I to maximise profit

$$\pi = (p_C - c)x + (p_B - c)y, \quad (7)$$

subject, if it remains unregistered ($I = 0$), to keeping sales $p_C x + p_B y$ below the registration threshold s^* .

Who gains from registering. Comparing optimised profit in (7) across $I = 1$ and $I = 0$, the net benefit of registering is, to first order, proportional to

$$t [\omega - \beta(1-\omega)]. \quad (8)$$

The two terms are the two faces of VAT. The reclaim benefit $t\omega$ rises in the input share ω : an input-heavy firm gains by recovering the VAT on its inputs. The output burden $t\beta(1-\omega)$ rises in the B2C sales share β : a consumer-facing firm loses, because the output

VAT it cannot pass on falls on its B2C sales. Hence (8) is *negative* for consumer-facing, low-input firms—for them the threshold is a downward notch and the model predicts bunching just below it—and *positive* for input-heavy, B2B-facing firms, who register *voluntarily* even below the threshold. This sign flip is the model’s central qualitative content, and it rationalises the administrative fact that some 43% of below-threshold UK firms register voluntarily (Liu et al., 2021). Figure 8 contrasts two representative firms—a consumer-facing, low-input firm and an input-heavy, business-facing one—and Figure 9 traces how the sign of the notch varies across firm types under the hard notch and a graduated taper.

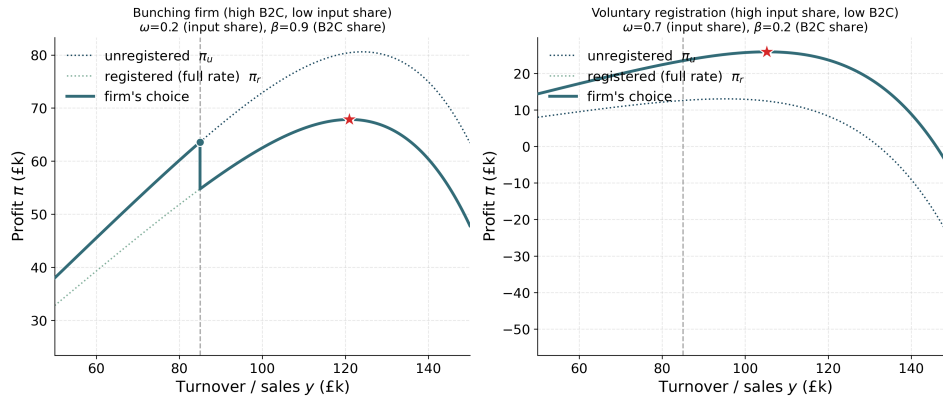


Figure 8: Illustrative. Net profit around the £85,000 threshold for two representative firms: each firm’s chosen profit (solid) against its unregistered and full-rate-registered branches (dotted). The consumer-facing, low-input firm ($\omega = 0.2$, $\beta = 0.9$) faces a downward notch at registration and bunches just below the threshold; the input-heavy, business-facing firm ($\omega = 0.7$, $\beta = 0.2$) gains from input reclaim and registers *voluntarily* even below it—the sign flip of (8). The (ω, β) values here are not yet data-grounded; in future work they are to come from ONS Supply–Use tables, including the final-demand split.

7.2 Parametric budget cost on the registration margin

This costing prices the *extensive* margin only—whether each firm sits inside or outside the VAT net—and holds turnover fixed. The *intensive* response embedded in the CES demand (5), by which registration depresses B2C quantity, is part of the model but is not re-solved here; pricing it is the full behavioural exercise left to future work. On the extensive margin, the registration decision (8) has a direct, static implication for revenue. Consider the VAT the government collects per firm. A *registered* firm remits VAT on its value added, $t(1 - \omega)y$, since it charges output VAT and reclaims input VAT. An *unregistered* firm remits nothing directly, but the input VAT it cannot reclaim, $t\omega y$, sticks and reaches the Exchequer as irrecoverable tax embedded in its purchases. A firm is inside the VAT net if it must register, $y \geq T^*$, or chooses to, $\omega - \beta(1 - \omega) > 0$. A reform

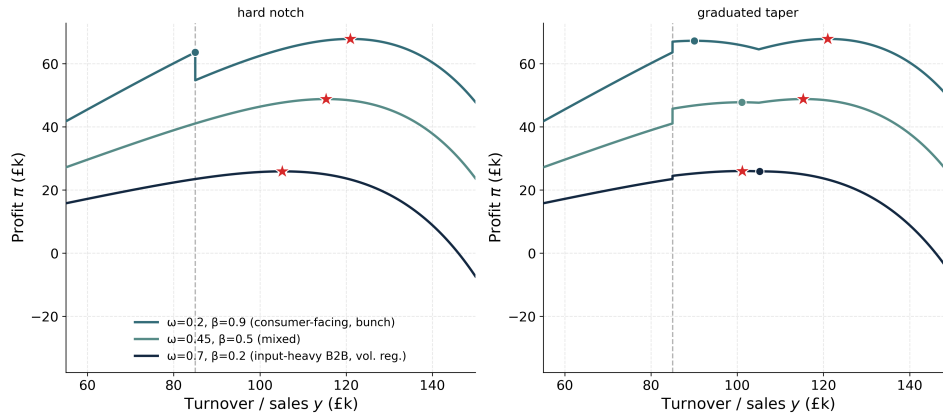


Figure 9: *Illustrative.* Net profit around the registration threshold for a consumer-facing firm under (7): the output-VAT burden on B2C sales creates a discrete drop at registration—a downward notch generating bunching just below s^* —which a smoothing taper would replace by a graduated slope. The underlying (ω, β) are illustrative and not yet grounded in ONS Supply–Use data.

that changes who is inside the net therefore moves revenue per affected firm between $t(1 - \omega)y$ (registered) and $t\omega y$ (unregistered)—a swing that depends entirely on whether ω is above or below $\frac{1}{2}$. Applying this accounting to the firm population, weighted by the assumed composition (ω, β) , yields a budget cost for each reform. Table 3 reports these population-weighted figures.

Table 3: Parametric static budget cost of the three threshold/rate reforms under the extensive-margin accounting of (8), by assumed firm composition (ω, β) (£bn; negative denotes a revenue loss). Each row fixes the input share ω and B2C sales share β for the affected population and applies the registered/unregistered remittance swing between $t(1 - \omega)y$ and $t\omega y$. Figures are conditional on the assumed (ω, β) and are a sensitivity range, not a forecast.

Firm type (ω, β)	Raise to £100k	Reduced 10% band	Reduced 15% band
Consumer-facing (0.2, 0.9)	−1.045	−0.933	−0.466
Mixed (0.45, 0.5)	0.000	−0.641	−0.321
Input-heavy B2B (0.7, 0.2)	0.000	−0.350	−0.175
Population-average (0.5, 0.5)	0.000	−0.583	−0.291

The key finding is that raising the threshold costs revenue *only* for the low- ω , high- β consumer-facing firms. These firms de-register when the threshold rises, and because $\omega < \frac{1}{2}$ for them they then yield less as unregistered firms ($t\omega y$) than they did as registered firms ($t(1 - \omega)y$), so the Exchequer loses. Input-heavy firms, by contrast, register *voluntarily* under (8) regardless of where the threshold sits, so a change in the threshold is irrelevant to them and the raise-to-£100k column is zero for every type with $\omega \geq \frac{1}{2}$.

Across the plausible range of (ω, β) the raise-to-£100k cost therefore ranges from £0 to about –£1.05bn. The two reduced-rate bands, which lower the output-VAT burden on *everyone* who would otherwise be registered rather than only the de-registering margin, cost more uniformly across firm types: up to about –£0.93bn for the 10% band and –£0.47bn for the 15% band, with the population-average composition (0.5, 0.5) implying –£0.58bn and –£0.29bn respectively. Figure 10 plots the reform cost against ω and the raise-to-£100k cost surface over the full (ω, β) plane.

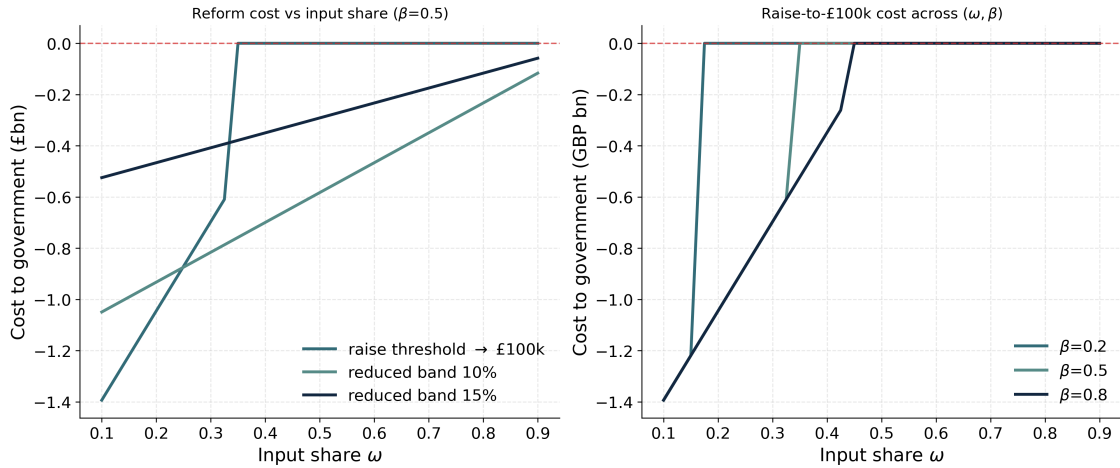


Figure 10: Parametric budget cost of the threshold reforms. Left: cost of each reform against the input share ω , showing that the raise-to-£100k cost is borne entirely by low- ω firms. Right: the raise-to-£100k cost surface over the (ω, β) plane, ranging from £0 (input-heavy or business-facing) to about –£1.05bn (consumer-facing, low-input). All magnitudes are conditional on the assumed composition (ω, β) and are read as a sensitivity range.

Caveats and scope. I am explicit about what this layer is and is not. It is a *static, extensive-margin* estimate: firms switch registration status, but their turnover and prices are held fixed. It is *not* the full behavioural response of Liu et al. (2021), in which the CES demand system (5) is re-solved through the optimal prices (6) so that prices, quantities, and the tax base all adjust jointly; recovering that, together with grounding ω and β in ONS Supply–Use data (including the final-demand split), is left for future work. The treatment of B2B VAT cascading is simplified: I credit unregistered firms only with their own irrecoverable input VAT and do not trace tax embedded further down the chain. And the (ω, β) values are not data-grounded, so the figures in Table 3 are a sensitivity range rather than point estimates or a forecast. This margin is exactly the “turnover-tax notch” corner of the model in Section 6: in the limit $\omega \rightarrow 0, \beta \rightarrow 1$ —an all-B2C firm with no reclaimable inputs—the registration choice collapses to the pure downward notch of Kleven and Waseem (2013), and the dominated region of Section 6 is the elasticity-free object that survives whatever one assumes about (ω, β) . That dominated region, and the static cost of Section 4, remain the paper’s robust anchors; the registration model adds a

Table 4: Summary of headline estimates. This table consolidates results reported and defined in the body sections cross-referenced below; it is a reference summary, not a new analysis. All quantities refer to the £85,000 data-year baseline. Static revenue figures are 2025–26.

Object	Symbol / definition	Value	Source
Static cost, £85k to £90k anchor	—	$\approx -£175\text{m}$	§4
Static sweep, £90k baseline	per £5k step	$\sim £150\text{--}220\text{m}$	Table 1
Dominated region (misallocation zone)	$a = T^*\tau/(1 - \tau)$	£21,250	§6.2
Firms in the dominated band (weighted)	—	$\approx 137,000$	§6.2
Registration-margin raise-to-£100k	parametric (ω, β) range	£0 to $-£1.05\text{bn}$	§7

disciplined account of *who* bears a threshold reform’s cost, conditional on a composition that future ONS work will pin down.

8 Conclusion

This paper contributes an open, reproducible firm-level microsimulation for analysing the UK VAT registration threshold, released in full—including its synthetic-data generator—for replication. Its value is as usable policy infrastructure rather than a structural identification result: I make no claim to recover policy-invariant firm primitives, unlike [Garicano et al. \(2016\)](#) and [Gourio and Roys \(2014\)](#). The analysis is anchored to the 2023–24 data year, in which the statutory threshold was £85,000. Table 4 collects the headline objects.

Findings. Four results stand out. First, the static pipeline reproduces HMRC’s costing of the £85,000 to £90,000 move to within about £10m (£175m against £185m)—a consistency check, since the microdata are calibrated to HMRC aggregates—and prices a menu of level and schedule reforms on a common £85,000/£183.6bn base: $-£508\text{m}$ to raise the threshold to £100,000, $-£336\text{m}$ for a graduated taper, and $-£343\text{m}$ and $-£171\text{m}$ for 10% and 15% banded reduced rates. Second, the notch’s distortion is captured exactly by its dominated region, the [Kleven and Waseem \(2013\)](#) width $a = T^*\tau/(1 - \tau) = £21,250$ —an accounting identity rather than a new structural result, but one spanning a populous range of roughly 137,000 firms—a range that fiscal drag widens further: with the £90,000 threshold now frozen in nominal terms, ageing the turnover distribution forward draws about 14% more firms into the dominated region by 2028–29 (roughly 130,000 to 148,000), with the just-below, bunching-exposed band growing about 9.5% (roughly 41,000 to 45,000). Setting cost against distortion is instructive: raising the threshold is the most expensive option yet merely relocates the zone, and a banded reduced rate does not shrink the distortion either—it lowers the largest single notch (from £17,000 to £12,750 at a 15% band) but, because the band reverts to the standard rate at its £105,000 top, adds a secondary notch, leaving the total dominated-turnover width

essentially unchanged (£21,562 at 15%, £22,569 at 10%, both marginally above the £21,250 baseline). Only the graduated taper, being continuous, removes the distortion entirely, and for less. Third, the open data reproduce the published density step at the threshold, but a placebo test shows the apparent excess mass is inherited from the coarse HMRC band targets rather than firm behaviour—it collapses from 8,712 to roughly 99 weighted firms when the target step is removed—so behavioural bunching is not identified on these data; the bunching ratio and implied elasticities are relegated to the appendix and labelled non-identified, and the behavioural evidence on threshold bunching remains the administrative estimate of [Liu et al. \(2021\)](#). Fourth, the framework also prices each reform *behaviourally* on the registration margin, embedding the extensive-margin registration decision in the framework of [Liu et al. \(2021\)](#): a firm weighs the input-VAT reclaim benefit, which rises in its input share ω , against the output-VAT burden on business-to-consumer sales, which rises in its B2C share β , so the net incentive to register is proportional to $t[\omega - \beta(1 - \omega)]$ (Section 7). The sign is informative: it is negative for consumer-facing, low-input firms, which have an incentive to remain unregistered (bunching), and positive for input-heavy business-to-business firms, which register voluntarily. I price each reform’s budget cost on this margin parametrically over (ω, β) —a registered firm yields $t(1 - \omega)y$ and an unregistered firm $t\omega y$ —reporting a sensitivity range rather than a point estimate. Raising the threshold to £100,000 costs between £0 and about –£1.05bn, the loss falling only on low- ω , high- β consumer-facing firms, because input-heavy firms register voluntarily and the threshold is irrelevant to them; the 10% and 15% reduced-rate bands cost up to about –£0.93bn and –£0.47bn. This is a static extensive-margin estimate over (ω, β) values that are not yet data-grounded, not the full CES-demand response of [Liu et al. \(2021\)](#). The static costs and the dominated region remain the robust, behaviour-free anchors; the registration-margin magnitudes are conditional on the assumed (ω, β) .

Limitations and future work. The model is a turnover-tax-notch approximation: it treats VAT as a cost on turnover, whereas real VAT is levied on value added with input reclaim, and roughly 43% of below-threshold firms register voluntarily ([Liu et al., 2021](#)), so the effective notch is smaller than a flat treatment implies. The dominated-region width indexes the misallocation distortion only and omits the compliance and administrative-cost saving that a *higher* threshold delivers by removing small firms from VAT—the classic rationale for a high registration threshold ([Keen and Mintz, 2004](#))—so the efficiency comparison is partial; a full welfare/MVPF accounting would net these compliance costs against the misallocation distortion. The menu reforms are also costed only in aggregate: because VAT liability by sector is excluded from the calibration optimiser (Section 3), the framework does not yet report the sectoral incidence of a taper or reduced rate—which sectors gain and by how much—a cut of obvious policy interest that I leave to future work. The microdata are synthetic and calibrated to coarse aggregate bands—which is why the registration-margin magnitudes are reported only as a parametric range over the input share ω and the B2C share β rather than as estimates—and the analysis is partial-equilibrium, abstracting from pass-through into prices ([Benedek](#)

et al., 2015; Bellon et al., 2022). The registration-margin costing is now done (Section 7), but the shares (ω, β) it requires are set parametrically rather than measured, and the response is the static extensive margin only; the chief outstanding tasks are therefore to ground (ω, β) in firm-level data—for instance the ONS Supply–Use tables—and to extend the static extensive margin to the full CES-demand response of Liu et al. (2021), which together would turn the present parametric range into estimates. A welfare/MVPF analysis modelling value added and input reclaim would in turn convert the dominated-region distortion into cost–benefit terms and support revenue-neutral schedule designs that neutralise the notch at no net fiscal cost. Above all, validation against administrative firm-level data with genuine location choice would replace the present internal-consistency checks with out-of-sample tests and let the framework identify firm bunching directly.

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A Appendix

This appendix collects supporting material in two parts: data construction and calibration robustness (Appendix A.1); and statistical inference for the reduced-form bunching estimates (Appendix A.2), which also reports a placebo test demonstrating that the synthetic bunching is mechanical—inherited from the coarse HMRC turnover-band targets—rather than behavioural.

A.1 Data construction detail

Section 3 describes the two-stage synthetic-population pipeline, parameterised by the registration threshold, and its calibration to HMRC and ONS aggregates. This appendix records the implementation detail deferred from that account.

Each of the approximately 2.94 million firm rows carries a turnover, an input expenditure, a calibration weight, and a registration status; the weights are chosen so the weighted totals reproduce the official targets while the population sums to about 2.5 million firms. Because the population is calibrated *to* the HMRC aggregates, agreement between my weighted outputs and those aggregates is a check of *internal consistency*, not external validation: any target used in calibration is reproduced by construction. I therefore read the $\sim 90\%$ calibration accuracy, and the close match of the static £85,000 $\rightarrow$ £90,000 costing to HMRC’s published figures, as consistency checks on the pipeline rather than out-of-sample tests of firm behaviour.

The reported calibration score is a bounded summary statistic. For each dimension I compute $\max\{0, 1 - |\hat{T} - T|/|T|\}$, so a dimension missed by more than 100 percent receives zero rather than a negative score. The headline “overall” number is the unweighted mean of the five dimensions used as calibration targets—HMRC turnover bands, ONS population, employment bands, HMRC sector counts, and VAT liability by turnover band. VAT liability by sector is excluded from that mean and reported separately as an informational diagnostic, because the current generator does not calibrate sector-specific input/output VAT structure.

Ledger and Populace migration check. The official target tables are being moved into PolicyEngine Ledger, with Populace providing the shared synthetic-population generator. I therefore keep the paper’s archived processed CSVs as the reproduction source for the reported results, and treat the pinned Populace/Ledger path as an auditable migration check rather than a silent replacement. The snapshot used here is PolicyEngine/ledger pull request 67 at commit cd98b5c and PolicyEngine/populace pull request 223 at commit fa20daf; both upstream pull requests merged on June 30, 2026. For the 2024–25 vintage, the Ledger-backed targets match the paper’s processed numeric inputs exactly after dropping presentation-only labels, totals, and the HMRC “Unknown” column that the generator does not calibrate: six normalized source tables checked, zero mismatches, and maximum numeric difference zero. A 1,000-iteration Populace run from those targets generated 2,946,015 firm rows. Its own validator reports a 93.8 percent overall

calibration score, compared with the paper validator’s 90.5 percent score for the same vintage, but this overall comparison is not like-for-like: HMRC turnover-band accuracy uses different band sets, and sector distribution reflects different calibration-target definitions. The directly comparable rows are ONS population, employment bands, and VAT liability by turnover band. Populace hits VAT liability by turnover band more closely, while its weighted population (2,945,777 versus 2,577,078) differs more from the paper population. VAT liability by sector remains an informational diagnostic in both runs. The comparison table and structured provenance are reproduced by `firm-microsim-populace-ledger` and checked into `results/populace_ledger_comparison.txt` and `results/populace_ledger_provenance.json`.

Counterfactual exclusion window and polynomial degree. The no-VAT counterfactual density of Section 5 is fitted by polynomial regression on the observed density outside a manipulation window around the threshold. The baseline excludes $[T^* - 15k, T^* + 15k]$ and fits a degree-7 polynomial. Widening the window guards against contamination of the counterfactual by the behavioural response but reduces the data available for the fit; raising the degree improves in-sample fit but risks over-fitting the tails. The sensitivity of the excess-mass and bunching-ratio estimates to window width and degree, with associated uncertainty, is reported in Appendix A.2.

A.2 Inference for the bunching estimator

This appendix documents the inference procedure for the reduced-form bunching estimates of Section 5: the mass-conservation constraint that disciplines the counterfactual density, the bootstrap that delivers standard errors, the sensitivity of the estimates to the polynomial degree, the exclusion window, and the effective wedge τ_e , and a placebo test demonstrating that the synthetic bunching is mechanical rather than behavioural. All numbers reported here and in the calibration step are produced by the released estimator (fixed seed, $B = 200$ bootstrap replications), run on the £85,000 (2023–24) weighted synthetic firm data; the seed and B are fixed so the table is reproducible, and nothing is hardcoded as a result.

Mass-conservation constraint

The canonical bunching estimator of Saez (2010), Kleven and Waseem (2013) and Chetty et al. (2011) imposes that the counterfactual density conserves mass: every firm removed from above the threshold reappears as excess mass below it. An unconstrained fit by plain polynomial OLS on bins outside a hand-set window computes the excess mass E (below T^*) and the missing mass Δ_R (above T^*) *independently*, with no guarantee that $E = \Delta_R$ and no endogenous location of the marginal buncher.

I impose the constraint in two ways. First, the counterfactual polynomial in $(y - T^*)$ is fit by an iterated weighted least-squares procedure in which the density of the bins above the excluded region is rescaled until the fitted counterfactual integrates to the *observed* total mass over the estimation range [£20k, £140k]. This forces the bunching mass to be

reabsorbed into the counterfactual rather than discarded. The regressor is normalised to $[-1, 1]$ for numerical stability of the high-degree Vandermonde basis. Second, the upper edge of the excluded (manipulation) region—the *marginal buncher* y_R —is located *endogenously* by integrating the missing mass upward from T^* until cumulative missing mass above equals cumulative excess mass below,

$$\int_{T^*}^{y_R} [f^{\text{cf}}(y) - f^{\text{obs}}(y)]_+ dy = \int_{T^*-W}^{T^*} [f^{\text{obs}}(y) - f^{\text{cf}}(y)]_+ dy = E,$$

rather than being fixed at an arbitrary window edge. The excess turnover span of the marginal buncher, $\Delta y^* = y_R - T^*$, is then used by the notch-width elasticity.

Imposing the constraint matters because the unconstrained fit grossly violates adding-up: it delivers $E = 7,820$ but $\Delta_R = 16,524$, whereas the constrained estimator returns $E = 8,712$ and $\Delta_R = 9,214$, which agree to within sampling noise. Imposing the constraint thus reconciles the excess mass below the threshold with the missing mass above it, which the unconstrained estimator computes independently and which would otherwise diverge by roughly a factor of two; the resulting movement in the headline estimates is reported in the calibration step.

Bootstrap standard errors

I compute standard errors by a firm-level bootstrap. In each of $B = 200$ replications I draw n firms with replacement from the estimation sample, with draw probabilities proportional to the survey weight, re-bin the resampled firms into £1,000 bins, and re-run the entire estimator—counterfactual fit, marginal-buncher location, and all downstream statistics—on the replicate. The standard error of each statistic is the standard deviation of its bootstrap distribution, and the 95% confidence interval is the 2.5th–97.5th percentile interval. Resampling firms (rather than residuals) propagates both sampling variation and the variation induced by re-binning. Table A.1 reports the results. Because these replications resample the ≈ 2.94 million *constructed* synthetic firm rows, whose number is itself a modelling choice, the resulting standard errors and 95% intervals quantify only the estimator’s dispersion under resampling of a synthetic file of analyst-chosen size; they scale with that arbitrary row count and should *not* be read as population sampling error about the universe of UK firms. They are reported for completeness and reproducibility, not as confidence statements about UK firms.

Non-identification: the elasticities are not behavioural

The bunching-implied elasticities in Table A.1—the notch-width elasticity (≈ 0.45) and the marginal-buncher elasticity (≈ 0.66)—together with the CES substitution elasticity $\sigma = 4.85$ are *not* behaviourally identified on these data. The synthetic firm population is calibrated to coarse aggregate HMRC turnover-band counts, and the generator contains no firm location-choice: firms do not respond to the threshold, so the excess mass below it is mechanical, inherited from the density step in the calibration targets

Table A.1: Bootstrap inference, £85,000 data ($B = 200$ firm resamples).

Statistic	Point	Std. error	95% CI
Bunching ratio b	0.060	0.005	[0.050, 0.068]
Excess mass E	8,712	662	[7,920, 10,516]
Substitution elasticity σ	4.85	0.12	[4.58, 5.08]
Displaced share Π	0.211	0.005	[0.200, 0.219]
Notch-width elasticity (median)	0.448	0.027	[0.405, 0.496]
Marginal-buncher elasticity	0.657	0.082	[0.510, 0.821]

rather than from any behavioural response (the placebo test below makes this precise). The elasticities are therefore reported only as a record of what the estimator returns when applied to these data; the paper draws no behavioural conclusion from them. A reader should note that the iso-elastic turnover elasticities $e \in \{0.05, 0.17, 0.32\}$ swept in the body (Section 7) are a *different* object from the notch-width elasticity (≈ 0.45) and marginal-buncher elasticity (≈ 0.66) reported here: the former are the median and mean of a per-firm marginal-buncher mapping under the iso-elastic firm specification, while the latter are aggregate estimator outputs computed under different normalisations. All of these numbers are mechanical artefacts of the calibration and are non-identified on these synthetic data—they admit no precise numerical mapping between the two sets—so none is used anywhere as a behavioural estimate. This is an instance of the general non-identification of bunching elasticities absent exogenous variation or strong functional-form assumptions (Blomquist et al., 2021; Bertanha et al., 2023).

Sensitivity to polynomial degree and exclusion window

Table A.2 maps the point estimates across a grid of polynomial degree $\in \{5, 6, 7, 8\}$ and symmetric exclusion window $\in \{\text{£}10\text{k}, \text{£}15\text{k}, \text{£}20\text{k}, \text{£}25\text{k}\}$. The estimates are stable across the central cells (degree 6–7, window £10–£20k): the substitution elasticity stays in roughly 4.4–5.7 and the excess mass in 4,000–9,000. The corner cells are less reliable: at very wide windows (£25k) the excluded region begins to swallow genuine curvature in the density, the fitted counterfactual crosses the observed density, and b can turn slightly negative (so σ is undefined); at the lowest degree (5) the inflexible trend over-fits the bunching mass into the counterfactual and inflates b . I therefore report the degree-7, £15k-window specification as the baseline, with the bootstrap CIs above spanning the variation across the stable interior of the grid.

Sensitivity to the effective wedge

The CES substitution elasticity is recovered as $\sigma = \ln(\text{RR}) / \ln(1 + \tau_e)$, where RR is the ratio of counterfactual-to-observed above/below mass ratios. Because the wedge enters only through $\ln(1 + \tau_e)$, σ scales roughly as $1 / \ln(1 + \tau_e)$ while the displaced share $\Pi = 1 - (1 + \tau_e)^{-\sigma}$ is essentially invariant (it is pinned by RR). On my data, $\tau_e = 0.025$ gives

Table A.2: Degree $\times$ window sensitivity (point estimates), £85,000 data. Cells with $b < 0$ leave σ undefined.

Degree	Window	b	E	σ	Notch-width (median)
5	£10k	0.150	13,339	5.62	0.47
5	£15k	0.139	18,823	5.16	0.71
5	£20k	0.141	25,596	4.95	0.87
5	£25k	0.152	34,172	5.11	1.04
6	£10k	0.077	7,262	5.42	0.41
6	£15k	0.045	6,575	4.99	0.44
6	£20k	0.015	3,822	4.21	0.35
6	£25k	-0.028	471	—	0.15
7	£10k	0.091	8,539	5.07	0.47
7	£15k	0.060	8,712	4.85	0.45
7	£20k	0.034	7,313	4.62	0.43
7	£25k	-0.004	3,089	—	0.25
8	£10k	0.118	10,806	4.92	0.42
8	£15k	0.118	16,329	4.53	0.59
8	£20k	0.166	29,364	4.96	0.77
8	£25k	0.301	59,931	7.57	0.95

$\sigma = 9.59$, $\tau_e = 0.05$ gives $\sigma = 4.85$, $\tau_e = 0.075$ gives $\sigma = 3.28$, and $\tau_e = 0.10$ gives $\sigma = 2.49$, while $\Pi = 0.211$ throughout. The magnitude of σ is therefore conditional on τ_e : I adopt $\tau_e = 0.05$ as a baseline effective registration wedge and report Π , which is robust to it, as the policy-relevant displacement object.

Placebo test

The non-identification of the bunching elasticities can be made precise with a placebo test. The HMRC turnover-band targets to which the synthetic population is calibrated are coarse: a single count for firms at or below £85,000 and a single count for firms in £85,000–£150,000. The implied average per-£1,000 densities differ sharply across the threshold ($\approx 7,981$ firms/£1,000 below versus $\approx 4,697$ above), so the calibration target *itself* contains a density step at exactly £85,000. Because the generator has no firm location-choice, any density discontinuity in the synthetic population must be inherited from this stepped target rather than produced by behaviour. The placebo removes the step and re-runs the estimator. Table A.3 reports two variants: Placebo A reweights the existing synthetic firms so the per-£1,000 density follows a single smooth log-quadratic trend across the threshold (preserving total mass over [£20k, £140k], no regeneration); Placebo B replaces the two HMRC band counts straddling £85,000 with a split implied by a single common per-£1,000 density (no step in the target), then re-runs the full generator and multi-objective calibration before estimating.

When the density step is removed from the calibration target, the estimated excess

Table A.3: Placebo test: removing the calibration density step collapses the excess mass. £85,000 data.

Population	b	E	b_{LLAT}
Actual (control)	+0.060	8,712	0.941
Placebo A (reweight density smooth)	−0.063	0	0.000
Placebo B (un-stepped HMRC band targets)	−0.016	99	0.010

Table A.4: Recovery test: the estimator recovers $\approx 90\%$ of an injected behavioural response of known magnitude. £85,000 data; $E_{\text{recovered}}$ is the change in signed below-threshold excess relative to the step-free baseline.

E_{true} (injected)	$E_{\text{recovered}}$	Recovery
2,000	1,792	89.6%
5,000	4,479	89.6%
8,000	7,167	89.6%

mass collapses from 8,712 firms in the control to at most ≈ 99 firms ($\approx 1\%$ of the control value) in Placebo B, and to zero in Placebo A; the bunching ratio even turns slightly negative. This proves that the synthetic bunching is inherited from the coarse HMRC turnover-band targets, not from any firm behaviour, and confirms that the elasticities in Table A.1 are mechanical artefacts of the calibration rather than behavioural parameters. The placebo is part of the released code and is fully reproducible.

Recovery: the estimator has power

The placebo establishes *specificity*: the estimator returns no bunching when none is present. The complementary property is *sensitivity*—whether the estimator can detect a genuine behavioural response when one *is* present. To check this I run a recovery exercise. Starting from the step-free Placebo A population—the null world with no bunching—I inject a behavioural bunching signal of *known* magnitude by relocating E_{true} firms, mass-conservingly, from a donor window just above the threshold to a window just below it (the firm location-choice the generator lacks), with a triangular profile peaking at £85,000. I then re-run the *same* estimator (degree 7, $\pm£15\text{k}$ window) and record the recovered excess mass, measured as the change in signed below-threshold excess relative to the step-free baseline. Table A.4 reports three magnitudes.

The estimator recovers about 90% of the injected mass at every magnitude (a stable 89.6%, monotone in E_{true} and never over-stating). The $\approx 10\%$ attenuation is benign: the degree-7 polynomial counterfactual re-absorbs part of the injected spike and the fixed exclusion window clips its tails, so the estimator slightly *under*-states a true response rather than inflating one. The estimator is therefore both *specific* (no false positive—the placebo) and *sensitive* (has power—this recovery). The implication is the one this section turns on: the absence of identified bunching on the actual synthetic data is a property

of the aggregate-calibrated *data*, which contain no firm location-choice, not a failure of the *method*. The recovery test is part of the released code and is fully reproducible.